LINUX / SYSTEMS

Linux Boot Process Visual Guide

UEFI/BIOS, GRUB, initramfs,
systemd stages, and boot failure triage.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Boot Timeline

Five acts from power button to login prompt.

- Firmware finds the bootloader (EFI System Partition, or MBR for legacy BIOS).
- GRUB loads the kernel + initramfs into memory and jumps to the kernel entry point.
- initramfs mounts the REAL root filesystem, then hands off to /sbin/init (systemd).
- systemd becomes PID 1 and walks sysinit → basic → multi-user → graphical.

STEP 2/6

UEFI vs BIOS & GRUB

Modern firmware, classic bootloader menu.

UEFI

GPT · ESP partition■Secure Boot capable

BIOS (legacy)

MBR · 2TB limit■no Secure Boot

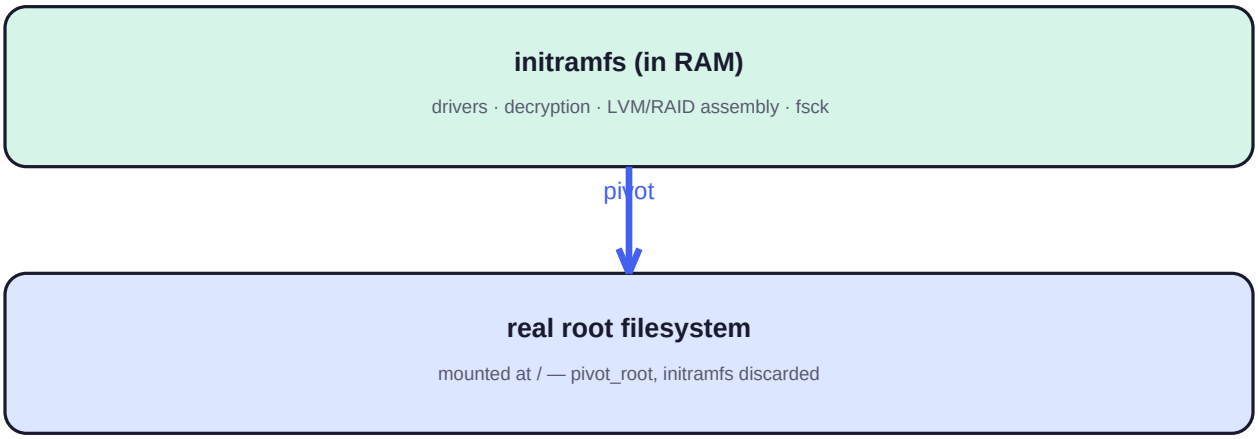
```
# GRUB menu → press 'e' to edit the boot entry:
linux /vmlinuz-6.8.0 root=/dev/sda2 ro quiet splash
# handy kernel parameters:
single        # boot to single-user (rescue)
init=/bin/bash  # skip systemd entirely (emergency shell)
nomodeset     # survive broken GPU drivers
```

init=/bin/bash is the skeleton key: it drops you to root without systemd, passwords, or mounts. Guard physical/console access accordingly — then remount -o remount,rw / to fix things.

STEP 3/6

initramfs: The Kernel's Starter Kit

A tiny filesystem with just enough to find the real one.

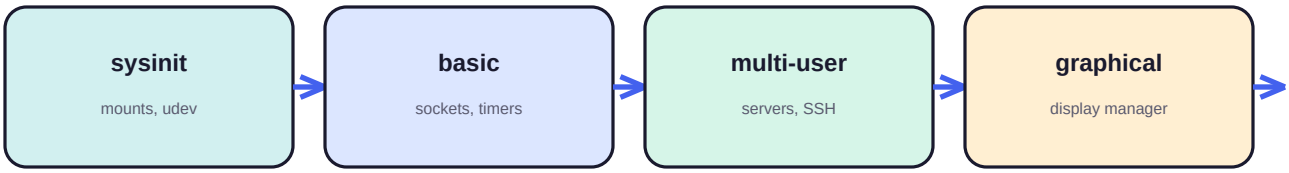


- Why it exists: the drivers for YOUR disk (NVMe, RAID, LUKS) can't live in the kernel image itself.
- Encrypted root? The passphrase prompt you see runs INSIDE initramfs, before / is mounted.
- Regenerate after driver/config changes: update-initramfs -u (Debian) / dracut (RHEL).
- A stale initramfs after adding a disk driver = 'can't find root device' at boot.

STEP 4/6

systemd Takes Over: Targets in Order

PID 1 climbs the milestone ladder.




```
$ systemd-analyze                    # total boot time
$ systemd-analyze blame | head       # slowest units
$ systemd-analyze critical-chain     # what the boot waited on
$ systemd-analyze plot > boot.svg    # visual timeline
```

critical-chain is the truth serum: it shows the dependency path that actually delayed boot, not just the slowest units in isolation.

STEP 5/6

Boot Failure Triage

Where it died tells you what broke.

Dies at	Likely cause	Recovery
no POST / firmware	hardware, boot order	BIOS setup, reseat, check ESP
GRUB rescue>	can't find config/disk	ls, set prefix+root, insmod normal
kernel panic	bad root=, driver, initramfs	boot older kernel; rebuild initramfs
can't mount root	LUKS/LVM/RAID/fstab	initramfs shell; check crypttab, mdadm
systemd emergency	failed fstab mount	fix /etc/fstab (nofail!), reboot

```
grub rescue> ls
grub rescue> set prefix=(hd0,gpt2)/boot/grub
grub rescue> insmod normal
grub rescue> normal
```

Add nofail to non-critical fstab entries (data disks, NFS): one missing disk then degrades the boot instead of dropping you to emergency mode.

STEP 6/6

Boot Hygiene Checklist

Boring habits that save 3 AM pages.

- ✓ Keep one older kernel installed — GRUB Advanced options is your rollback
- ✓ nofail on every non-root fstab entry (data, NFS, USB)
- ✓ Rebuild initramfs after storage/crypto driver changes — then TEST reboot
- ✓ Document LUKS passphrases and mdadm config somewhere off the machine
- ✓ systemd-analyze blame after major changes — catch slow boots early
- ✓ Test the rescue path BEFORE you need it: boot the old kernel once

STEP ✓I✓

Key Takeaways

Linux Boot Process

- Five acts: firmware → GRUB → kernel → initramfs → systemd (PID 1).
- initramfs carries the drivers needed to find the real root — encrypted, RAID, LVM.
- GRUB's edit mode (init=/bin/bash, single) is the emergency toolkit — guard console access.
- systemd climbs sysinit → basic → multi-user → graphical; analyze with critical-chain.
- Triage by death point; nofail in fstab and a spare kernel turn disasters into reboots.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev